Apple Lossless

Also known as Apple Lossless Encoder, ALE, or Apple Lossless Audio Codec, ALAC, this is an audio codec developed by Apple for lossless compression of digital music. Apple Lossless data is stored within an MP4 container with the filename extension .m4a. Apple claims that audio files compressed with its lossless codec will use up "about half the storage space" that the uncompressed data would require. Testers have found that compressed files are about 60 per cent the size of the originals, similar to other lossless formats. Compared to most other formats, Apple Lossless is not as difficult to decode, making it practical for a limited-power device such as an iPod.

FLAC

An acronym for Free Lossless Audio Codec, this is a popular format for audio compression. It is suitable both for everyday playback and for archiving audio collections. It was developed by the Xiph.Org Foundation (www.xiph.org), and is royalty-free.

WMA Lossless

Windows Media Audio actually is the name of Microsoft's solution for digital audio. WMA codecs once were only lossy, but with the release of Windows Media Encoder 9 Series in early 2003, Microsoft provides the option of lossless compression by Windows Media Audio 9 Lossless codecs. WMA Lossless is, formally, a digital audio file format that compresses an audio CD to a range of 206 to 411 MB, at bitrates of 470 to 940 kbps. It uses the same .WMA file extension as other Windows Media Audio formats.

2.5 Compressed Lossy Formats

Such formats permanently discard data to achieve reduction in file sizes. But to most ears, music in such formats is good enough for everyday listening.

MP3

A number of techniques are employed in MP3 to determine which